

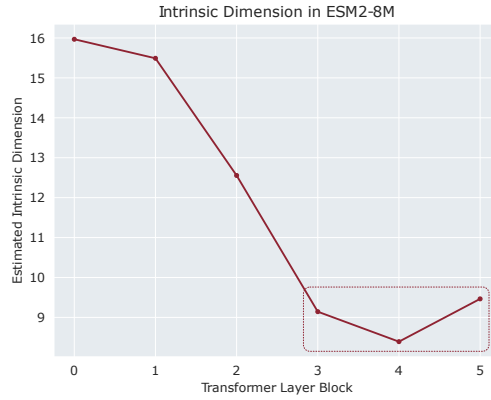


Figure 4: Evolution of the intrinsic dimension estimate through the layers of the ESM-2 8M model. We highlight the layers in the plateau/final ascent region.

A.3.2 SPARSE AUTOENCODER SELECTION

We evaluate all versions of the trained autoencoder primarily on two metrics: cross-entropy increase and sparsity (measured by L_0). These two goals are in conflict with each other, so we select what we think is a good compromise at the bend of the Pareto frontier (figure 5). The selected autoencoder has an average L_0 per amino acid of 18, and a cross-entropy increase of 0.10, with a hidden size that is 10 times larger than the original hidden size of the ESM-2 model. The number of dead latents in this SAE is 573. Decreasing the cross-entropy even more would entail a significant increase in activation density, which we want to avoid.

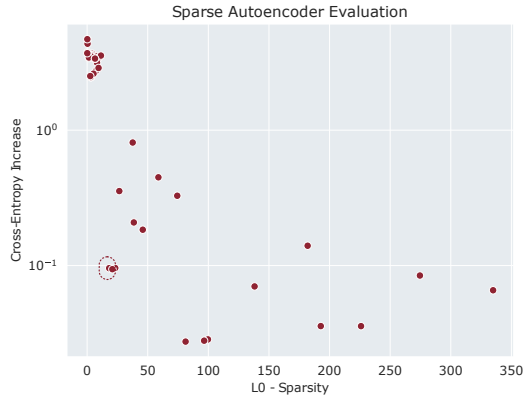


Figure 5: Cross-entropy increase vs sparsity trade-off for all the vanilla sparse autoencoders trained on layer 3 embeddings from ESM-2 8M. The selected autoencoder is indicated by a dashed circle.

A.3.3 INTERPRETING LATENTS - ACTIVATION PLOTS

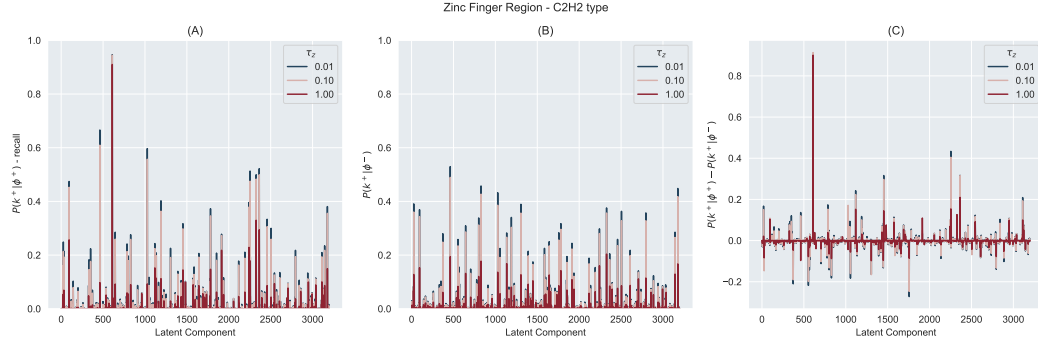


Figure 6: (A) $P(k^+|\phi^+)$ - Percentage of tokens for which each latent component is active when there is a C2H2 zinc finger type label, (B) $P(k^+|\phi^-)$ - percentage when there is no C2H2 zinc finger label, and (C) difference between these two values for three different activation thresholds τ_k . From the last plot, we see that there is a latent component that is prominently associated with the C2H2 zinc finger label.